



PAK-AUSTRIA FACHHOCHSCHULE

INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY



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Program : Cybersecurity

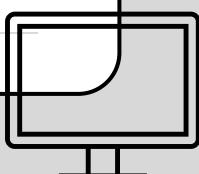
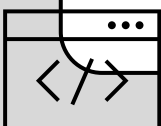
Instructor's name: Hassan Ali

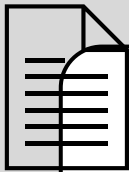
Date Submitted : 20 / 07 / 26

Deadline : 20 / 07 / 26

Assignment Report # 8

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• INTRODUCTION •

In this assignment, we explored the concepts of Active and Passive Reconnaissance and the tool required to perform them through a few tasks. The objective was to understand how these actions differ from each other, and they worked on their own. The assignment consists of three tasks; it goes as follows

Assignments Tasks:

Task #01 : *Passive Reconnaissance*, Select an authorized domain or intentionally vulnerable practice target. Perform passive reconnaissance using any three of the following tools: **WHOIS, theHarvester, Assetfinder**. Collect the following information:

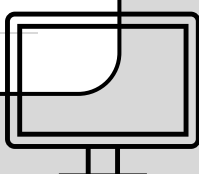
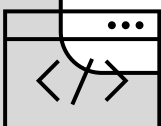
- Domain registration details
- DNS records
- Subdomains
- Email addresses
- Public IP addresses

Task #02 : *Active Reconnaissance*, Select an authorized target or a local lab environment. Perform active reconnaissance using at least three of the following tools: **Nmap, WhatWeb, Gobuster**. Record the following information:

- Open ports
- Running services
- Service versions
- Operating system (if detected)
- Web technologies
- Discovered directories or files.

Task #03 : *Comparison of Active and Passive*, Understand the differences, advantages, and limitations of active and passive reconnaissance techniques,

- Create a **Comparison Table**, Include the following: **Definition, Objectives, Advantages/Disadvantages, Common tools, Typical use cases**,
- Select five tools and describe: **Purpose, Basic command/example usage, Information gathered**,
- conclude by explaining when active reconnaissance should be preferred over passive reconnaissance and vice versa.



• TASK # 01 •

Task #01 : *Passive Reconnaissance*, Select an authorized domain or intentionally vulnerable practice target. Perform passive reconnaissance using three of the following tools: **WHOIS**, **theHarvester**, **Assetfinder**. Collect the following information:

Identify :

- Domain registration details
- DNS records
- Subdomains
- Email addresses
- Public IP addresses

I will be performing both actions on <http://65.61.137.117> Website.

Tool WHOIS :

The tool WHOIS Queries public databases to retrieve domain registration details, owner information, and IP address ownership. Following is the screen shot of the output,

```
# start
NetRange: 65.61.128.0 - 65.61.191.255
CIDR: 65.61.128.0/18
NetName: RSPC-NET-4
NetHandle: NET-65-61-128-0-1
Parent: NET-65-61-128-0-0
NetType: Direct Allocation
OriginAS:
Organization: Rackspace Hosting (RACKS-8)
RegDate: 2002-11-01
Updated: 2012-02-24
Ref: https://rdap.arin.net/registry/ip/65.61.128.0

OrgName: Rackspace Hosting
OrgId: RACKS-8
Address: 1718 Dry Creek Way
Address: Ste 115
City: San Antonio
StateProv: TX
PostalCode: 78259-1837
Country: US
RegDate: 2010-03-29
Updated: 2020-06-10
Ref: https://rdap.arin.net/registry/entity/RACKS-8

OrgTechHandle: ZR9-ARIN
OrgTechName: Rackspace, com
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/ZR9-ARIN

OrgAbuseHandle: ABUSE45-ARIN
OrgAbuseName: Abuse Desk
OrgAbusePhone: +1-210-312-4000
OrgAbuseEmail: abuse@rackspace.com
OrgAbuseRef: https://rdap.arin.net/registry/entity/ABUSE45-ARIN

OrgTechHandle: HANSE157-ARIN
OrgTechName: Hansell, Chris
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/HANSE157-ARIN

OrgNOCHandle: HANSE157-ARIN
OrgNOCName: Hansell, Chris
OrgNOCPhone: +1-210-312-4000
OrgNOCEmail: hostmaster@rackspace.com
OrgNOCRef: https://rdap.arin.net/registry/entity/HANSE157-ARIN

OrgTechHandle: IPADM17-ARIN
OrgTechName: IPADMIN
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/IPADM17-ARIN

# end

# start
NetRange: 65.61.137.64 - 65.61.137.127
CIDR: 65.61.137.64/26
NetName: RACKS-8-189343775333749
NetHandle: NET-65-61-137-64-1
Parent: RSPC-NET-4 (NET-65-61-128-0-1)
NetType: Reassigned
OriginAS:
Customer: Rackspace Backbone Engineering (C05762718)
RegDate: 2015-06-08
Updated: 2015-06-08
Ref: https://rdap.arin.net/registry/ip/65.61.137.64

CustName: Rackspace Backbone Engineering
Address: 9725 Datapoint Drive, Suite 100
City: San Antonio
StateProv: TX
PostalCode: 78229
Country: US
RegDate: 2015-06-08
Updated: 2015-06-08
Ref: https://rdap.arin.net/registry/entity/C05762718

OrgTechHandle: ZR9-ARIN
OrgTechName: Rackspace, com
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/ZR9-ARIN

OrgAbuseHandle: ABUSE45-ARIN
OrgAbuseName: Abuse Desk
OrgAbusePhone: +1-210-312-4000
OrgAbuseEmail: abuse@rackspace.com
OrgAbuseRef: https://rdap.arin.net/registry/entity/ABUSE45-ARIN

OrgTechHandle: HANSE157-ARIN
OrgTechName: Hansell, Chris
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/HANSE157-ARIN

OrgNOCHandle: HANSE157-ARIN
OrgNOCName: Hansell, Chris
OrgNOCPhone: +1-210-312-4000
OrgNOCEmail: hostmaster@rackspace.com
OrgNOCRef: https://rdap.arin.net/registry/entity/HANSE157-ARIN

OrgTechHandle: IPADM17-ARIN
OrgTechName: IPADMIN
OrgTechPhone: +1-210-312-4000
OrgTechEmail: hostmaster@rackspace.com
OrgTechRef: https://rdap.arin.net/registry/entity/IPADM17-ARIN

# end
```

Task # 01: WHOIS Output

Tool theHarvester :

The tool theHarvester gathers emails, subdomains, and hosts related to a target using public sources like search engines and PGP servers. Following is the screen shot of the output,

```
Session Actions Edit View Help
*****
[*] Target: 65.61.137.117

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: '65.61.137.117'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: '65.61.137.117'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: '65.61.137.117'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: '65.61.137.117'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for DNSDumpster.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dymo:
[!] Missing API key for dymo.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in fullhunt:
[!] Missing API key for fullhunt.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Error initializing SearchGithubCode:
[!] Missing API key for Github.
A Missing Key error occurred in github-code:
[!] Missing API key for Github.
```

```
Session Actions Edit View Help
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for HaveIBeenPwned.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Hunter:
[!] Missing API key for Hunter.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Hunter How:
[!] Missing API key for hunterhow.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in intelx:
[!] Missing API key for Intelx.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[*] Mojeek: No API key found, using default scraping mode.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Netlas:
[!] Missing API key for netlas.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Unexpected error occurred in Onyphe module:
[!] Missing API key for onyphe.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in PentestTools search:
[!] Missing API key for PentestTools.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in ProjectDiscovery:
[!] Missing API key for ProjectDiscovery.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in RocketReach:
[!] Missing API key for RocketReach.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for SecurityScorecard.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred Security Trails:
[!] Missing API key for SecurityTrail.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in sherlockkey:
[!] Missing API key for sherlockkey.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Shodan search:
[!] Missing API key for Shodan.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in Tomba:
[!] Missing API key for Tomba Key and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in venacius search:
```

```
Session Actions Edit View Help
[!] Missing API key for Venacius.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in virustotal search:
[!] Missing API key for virustotal.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in whoisxml search:
[!] Missing API key for whoisxml.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in zoomeye:
[!] Missing API key for zoomeye.
[*] Searching Baidu.
Chaos API error: get access token here https://cloud.projectdiscovery.io
[*] Searching Chaos.
Searching results.
[*] Searching Certspotter.
[*] Searching Duckduckgo.
Fofa API error: [-700] 账号无效
Failed to parse Fofa response:
[!] Missing API key for Fofa API (Invalid credentials).
[*] Searching Fofa.
[*] Searching Commoncrawl.
[*] Searching Hackertarget.
2026-07-19 16:45:08,430 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson Rock processing for: 65.61.137.117
2026-07-19 16:45:08,431 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson Rock search for: 65.61.137.117
2026-07-19 16:45:15,057 - theHarvester.discovery.hudsonrocksearch - INFO - Domain statistics: 1 total compromised, 0 employees, 1 users
2026-07-19 16:45:16,059 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock search completed. Found 0 hosts, 0 IPs, 0 emails
2026-07-19 16:45:16,059 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock processing completed successfully: 0 hosts, 0 IPs, 0 emails, 0 stealers
[*] Searching Hudsonrock.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
LeakIX API requires authentication: "Incorrect API Key"
[*] Searching CRTsh.
[*] Searching Gitlab.
[*] Searching Leaklookup.
[*] Searching Mojeek.
[*] Searching Otx.
[*] Searching LeakIX.
[*] Searching ShodanInternetDB.
[*] Searching Rapidids.
[*] Searching Robtex.
[*] Searching Thc.
No response from ThreatCrowd API for: 65.61.137.117
[*] Searching Threatcrowd.
[*] Searching Subdomainfinder99.
[*] Searching Uriscan.
[*] Windvane API key not found. Using limited unauthenticated access.
[*] API unavailable, using fallback subdomain pattern search...
[*] Searching Subdomaincenter.
```



```
[*] Searching Waybackarchive.
[*] Searching Yahoo.
[*] No additional subdomains found via fallback methods
[*] Searching Windvane.

[*] ASNS found: 1
AS33070

[*] Interesting Urls found: 3
http://65.61.137.117
http://65.61.137.117/login.jsp
http://65.61.137.117:443/

[*] No LinkedIn users found.

[*] LinkedIn Links found: 0

[*] IPs found: 2
65.61.137.117

[*] No emails found.

[*] No people found.

[*] No hosts found.

[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
```

Task # 01: theHarvester Output

Tool Assestfinder :

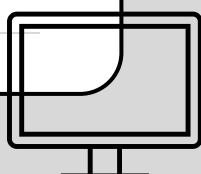
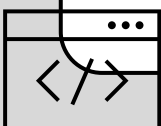
The tool Assestfinder Quickly discovers subdomains associated with a given domain using multiple online sources. Following is the screen shot of the output,

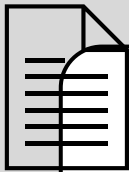
```
(kali㉿kali)-[~]
└─$ assetfinder 65.61.137.117
65.61.137.117

(kali㉿kali)-[~]
└─$

(kali㉿kali)-[~]
└─$ assetfinder demo.testfire.net
demo.testfire.net
click.myownrecepeworld.in
```

Task # 01: Assestfinder Output





Identified :

- **Domain registration details :**
 - Hosted by Rackspace Hosting
 - In San Antonio, Texas, USA.
 - net range 65.61.128.0 - 65.61.191.255.
- **DNS records :** demo.testfire.net
- **Subdomains :**
 - demo.testfire.net
 - click.myownreceipeworld.in
- **Email addresses :** Not Discovered.
- **Public IP addresses :** 65.61.137.117

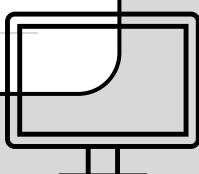
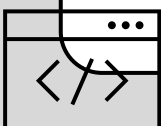
Task Conclusion :

In this Task we successfully performed passive reconnaissance and found out publicly available details.

• TASK # 02 •

Task #02 : Active Reconnaissance, Select an authorized target or a local lab environment. Perform active reconnaissance using at least three of the following tools: **Nmap, WhatWeb, Gobuster.** Record the following information:

- Open ports
- Running services
- Service versions
- Operating system (if detected)
- Web technologies
- Discovered directories or files.



Tool Nmap :

The tool Nmap is a powerful network scanner used to discover open ports, running services, service versions, and operating systems on a target. Following is the screen shot of the output,

```
(kali@kali)-[~]
$ nmap -sV -O 65.61.137.117
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-19 17:01 -0400
Nmap scan report for 65.61.137.117
Host is up (0.22s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache Tomcat/Coyote JSP engine 1.1
443/tcp   open  ssl/http  Apache Tomcat/Coyote JSP engine 1.1
8080/tcp   open  http      Apache Tomcat/Coyote JSP engine 1.1
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: DD-WRT v24-sp2 (Linux 2.4.37) (97%), Microsoft Windows XP SP3 or Windows 7 or Windows Server 2012 (97%), VMware Player virtual NAT device (97%), Actiontec MI24WR-GEN11 WAP (98%), Microsoft Windows XP SP3 (94%), Linux 3.2 (93%), Linux 4.4 (93%), BlueArc Titan 2100 NAS device (89%)
No exact OS matches for host (test conditions non-ideal).
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 95.34 seconds
```

Task # 02: Nmap Output

Tool WhatWeb :

The tool WhatWeb identifies web technologies, content management systems, servers, and frameworks used by a website. Following is the screen shot of the output,

```
(kali@kali)-[~]
$ whatweb 65.61.137.117
http://65.61.137.117 [200 OK] Apache, Cookies[JSSESSIONID], Country[UNITED STATES][US], HTTPServer[Apache-Coyote/1.1], HttpOnly[JSSESSIONID], IP[65.61.137.117], Java, Title[Alto ro Mutual]
https://65.61.137.117 [200 OK] Apache, Cookies[JSSESSIONID], Country[UNITED STATES][US], HTTPServer[Apache-Coyote/1.1], HttpOnly[JSSESSIONID], IP[65.61.137.117], Java, Title[Alto ro Mutual]
```

Task # 02: WhatWeb Output

Tool GoBuster :

The tool GoBuster is a fast tool used for brute-forcing and discovering hidden directories, files, and subdomains on web servers. Following is the screen shot of the output,

```
(kali@kali)-[~]
$ gobuster dir -u http://65.61.137.117 -w /usr/share/wordlists/dirb/common.txt -t 50

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

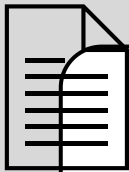
[+] Url: http://65.61.137.117
[+] Method: GET
[+] Threads: 50
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

admin (Status: 302) [Size: 0] [→ /login.jsp]
bank (Status: 302) [Size: 0] [→ /login.jsp]
aux (Status: 200) [Size: 0]
com3 (Status: 200) [Size: 0]
con (Status: 200) [Size: 0]
com2 (Status: 200) [Size: 0]
com1 (Status: 200) [Size: 0]
images (Status: 302) [Size: 0] [→ /images/]
lpt1 (Status: 200) [Size: 0]
lpt2 (Status: 200) [Size: 0]
nul (Status: 200) [Size: 0]
[ERROR] error on word fuckoff: timeout occurred during the request
[ERROR] error on word fuck: timeout occurred during the request
[ERROR] error on word fuckyou: timeout occurred during the request
pr (Status: 302) [Size: 0] [→ /pr/]
prn (Status: 200) [Size: 0]
static (Status: 302) [Size: 0] [→ /static/]
Progress: 4217 / 4613 (91.42%) [ERROR] error on word porn: timeout occurred during the request
util (Status: 302) [Size: 0] [→ /util/]
Progress: 4613 / 4613 (100.00%)

Finished
```

Task # 02: GoBuster Output

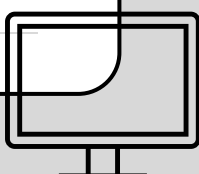
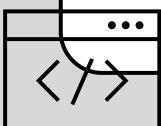


Identified :

- **Open ports :**
 - 80 (HTTP)
 - 443 (HTTPS)
 - 8080 (HTTP)
- **Running services :**
 - HTTP Web Server
 - SSL/TLS service on port 443
- **Service versions :** Apache Tomcat / Coyote JSP engine 1.1
- **Operating system :** Not Found.
- **Web technologies :**
 - Apache-Coyote/1.1
 - Java
 - JSESSIONID cookies
- **Discovered directories or files :**
 - /admin (redirects to login)
 - /bank (redirects to login)
 - /images

Task Conclusion :

In this Task we successfully performed active reconnaissance and found out services, web technologies and other details.



• TASK # 03 •

Task #03 : Comparison of Active and Passive, Understand the differences, advantages, and limitations of active and passive reconnaissance techniques,

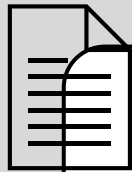
- Create a **Comparison Table**, Include the following: **Definition, Objectives, Advantages/Disadvantages, Common tools, Typical use cases**,
- Select five tools for both and describe: **Purpose, Basic command/example usage, Information gathered**,
- conclude by explaining when active reconnaissance should be preferred over passive reconnaissance and vice versa.

Comparison : Active and Passive Reconnaissance

The Following table describes the difference between Active and Passive Reconnaissance,

Point	Passive	Active
Definition	Gathering info without directly interacting with the target	Directly interacting with the target to gather info
Objectives	Collect public data quietly	Discover live services, ports, vulnerabilities
Advantages	Undetectable, legal in most cases, no risk of alerting target	More accurate, detailed, faster results
Disadvantages	Limited data, outdated info	Can be detected, may be illegal without permission, noisy
Common Tools	WHOIS, theHarvester, Assetfinder, Amass (passive), Subfinder	Nmap, Gobuster, Nikto, WhatWeb, RustScan
Use Cases	Initial research, OSINT, before engagement	During penetration test (with permission)

Task # 03: Difference Between Active and Passive Reconnaissance



Tools for Passive Reconnaissance

Tool # 01 : WHOIS

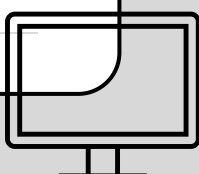
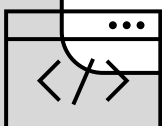
- **Purpose** : Retrieves public domain registration and IP ownership information from WHOIS databases.
- **Basic command** : `whois <target>`
- **Example** : `whois 65.61.137.117` OR `whois demo.testfire.net`
- **Information gathered**: Domain/IP registration details, owner organization, registration dates, abuse contacts, and network ranges.

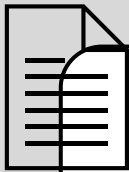
Tool # 02 : theHarvester

- **Purpose** : Collects emails, subdomains, and hosts using public search engines and data sources passively.
- **Basic command** : `theHarvester -d -b all`
- **Example** : `theHarvester -d demo.testfire.net -b all -l 500`
- **Information gathered**: Emails, subdomains, public IPs, and interesting URLs.

Tool # 03 : Assestfinder

- **Purpose** : Discovers subdomains associated with a domain using various online sources.
- **Basic command** : `assetfinder <domain>`
- **Example** : `assetfinder demo.testfire.net`
- **Information gathered**: Subdomains linked to the target.





Tool # 04 : Subfinder

- **Purpose** : Fast passive enumeration of subdomains using multiple public data sources and APIs.
- **Basic command** : subfinder -d target
- **Example** : subfinder -d demo.testfire.net -all -o subdomains.txt
- **Information gathered**: Valid subdomains discovered from public sources

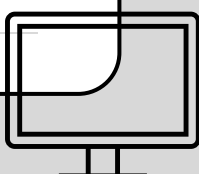
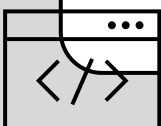
Tool # 05 : Amass (passive)

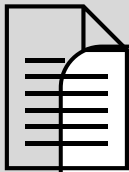
- **Purpose** : In-depth attack surface mapping using only passive OSINT sources for subdomains and related assets.
- **Basic command** : amass enum -passive -d target
- **Example** : amass enum -passive -d demo.testfire.net -o amass.txt
- **Information gathered**: Subdomains, related domains, ASNs, and infrastructure relationships from dozens of public sources.

Tools for Active Reconnaissance

Tool # 01 : Nmap

- **Purpose** : Powerful network scanner for discovering open ports, running services, service versions, OS details, and basic vulnerabilities.
- **Basic command** : nmap -sV -O target
- **Example** : nmap -sV -O 65.61.137.117 or nmap -sS -sV -O --script vuln 65.61.137.117
- **Information gathered**: Open ports , services ,service versions, OS guesses and scripts output.





Tool # 02 : WhatWeb

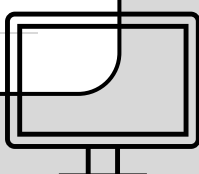
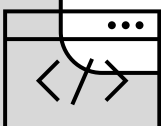
- **Purpose** : Identifies web technologies, servers, frameworks, and other details by analysing HTTP responses.
- **Basic command** : whatweb target
- **Example** : whatweb 65.61.137.117 or whatweb -v http://65.61.137.117
- **Information gathered**: Web server, technologies ,cookies , country, and page title .

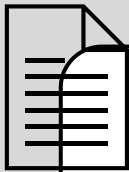
Tool # 03 : GoBuster

- **Purpose** : Brute-forces and discovers hidden directories, files, and subdomains on web servers.
- **Basic command** : gobuster dir -u url -w wordlist
- **Example** : gobuster dir -u http://65.61.137.117 -w /usr/share/wordlists/dirb/common.txt -t 50
- **Information gathered**: Directories and files.

Tool # 04 : Nikto

- **Purpose** : Web server scanner that checks for dangerous files, outdated software, misconfigurations, and potential vulnerabilities.
- **Basic command** : nikto -h target
- **Example** : nikto -h http://65.61.137.117 or nikto -h 65.61.137.117 -p 80,443
- **Information gathered**: Server issues, risky files/directories, software versions, and security misconfigurations.





Tool # 05 : RustScan

- **Purpose** : Extremely fast port scanner to quickly identify open ports.
- **Basic command** : `rustscan -a target`
- **Example** : `rustscan -a 65.61.137.117 -- -sV -O`
- **Information gathered**: Open ports and initial service detection at high speed.

When Active VS Passive Reconnaissance

Prefer Passive Reconnaissance when :

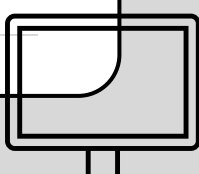
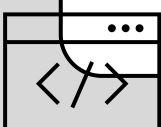
- You need to stay completely undetected (e.g., initial phase, red teaming, or when authorization is limited).
- You want zero risk of alerting the target or triggering security systems.
- You are in the early stages of reconnaissance to build a broad picture of the target (domain ownership, subdomains, emails, public infrastructure).
- Legal or ethical constraints are strict, passive recon is generally legal without explicit permission because it uses only public data.

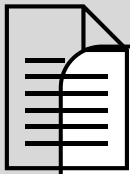
Prefer Active Reconnaissance when :

- You need accurate, up-to-date, and detailed information (exact open ports, service versions, running technologies, hidden directories).
- You have explicit written authorization (required for penetration testing or authorized assessments).
- The goal is to identify exploitable vulnerabilities, misconfigurations, or live attack surfaces.
- Speed and confirmation of findings are critical.

Task Conclusion :

In this Task we learned about the difference between active and passive reconnaissance, five tools for each and when should one be preferred over another.





Assignment Conclusion:

In this assignment we Learned about Active and Passive Reconnaissance, the difference between them, the tools required perform each of them, the details of the tools and their uses and when should one be preferred over another. This increased and polished my ability to use and learn more about cybersecurity.

• The End •

